

8 (**short**) A puck of mass m is sliding on an air table as shown in Fig. 5. It is connected by a light inextensible string to a mass $2m$ hanging below the table. The string passes through a frictionless hole in the table. The motion of the puck is described by its distance r from the hole and its angular velocity ω .

(a) The puck is moving in steady circular motion at radius r . Find an expression for the speed of the puck. [4]

(b) An impulse tangent to its motion is now delivered to the puck. The effect of this impulse is to send the puck to a maximum distance from the hole of $2r$. During this motion find the minimum speed of the puck. [6]

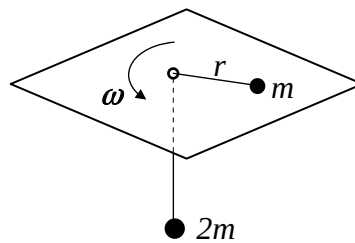


Fig. 5